

8086 Instruction Set Architecture

Q. Classify the Instruction Set of the 8086 microprocessor. Briefly explain Data Transfer, Arithmetic, Logical, Branch, String, and Processor Control instructions with one example each.

📌 Definition to Remember

An **Instruction Set** is the complete collection of commands a microprocessor can understand and execute. The 8086 Instruction Set tells the CPU how to manipulate data in registers/memory, perform math, and control the flow of execution.

1. Classification of Instructions

A. Data Transfer Instructions

Moves data between registers or between memory and registers. (Does NOT affect flags).

* **Example:** `MOV AX, BX` (Copies the 16-bit content of BX into AX).

* **Others:** `PUSH`, `POP`, `IN`, `OUT`, `XCHG`.

B. Arithmetic Instructions

Performs mathematical operations. (Updates the Flag register).

* **Example:** `ADD AL, 05H` (Adds 05H to AL and stores the result in AL).

* **Others:** `SUB`, `MUL`, `DIV`, `INC` (Increment), `DEC` (Decrement).

C. Logical Instructions

Performs bitwise boolean operations and bit shifting.

* **Example:** `AND CX, DX` (Performs bitwise AND on CX and DX; stores result in CX).

* **Others:** `OR`, `NOT`, `XOR`, `SHL` (Shift Left), `ROR` (Rotate Right).

D. Branch / Control Transfer Instructions

Changes the normal sequential flow of execution by altering the Instruction Pointer (IP).

* **Unconditional Example:** `JMP 1200H` (Jumps directly to memory address 1200H).

* **Conditional Example:** `JZ LABEL` (Jump to LABEL if the Zero Flag is set).

* **Others:** `CALL`, `RET`, `LOOP`.

E. String Instructions

Optimized to manipulate blocks of data (strings/arrays) in memory.

* **Example:** `MOVSB` (Moves a string byte from Source Index (SI) to Destination Index (DI)).

* **Others:** `CMPSB` (Compare string), `SCASB` (Scan string).

F. Processor Control Instructions

Controls processor state or hardware synchronization.

* **Example:** `STC` (Sets the Carry Flag to 1).

* **Others:** `CLC` (Clear Carry), `CLI` (Disable interrupts), `HLT` (Halt).

★ Must-Write Points (for 10 marks)

1. An **Instruction Set** defines all commands the 8086 CPU can execute.
 2. **Data Transfer:** Moves data (e.g., `MOV AX, BX`). Does not alter flags.
 3. **Arithmetic:** Math operations (e.g., `ADD AL, 05H`). Updates flags.
 4. **Logical:** Bitwise operations (e.g., `AND CX, DX`, `SHL`).
 5. **Branch/Control:** Alters execution flow (e.g., `JMP`, `JZ`). Changes the IP register.
 6. **String:** Operates on blocks of memory using SI and DI (e.g., `MOVSB`).
 7. **Processor Control:** Manages hardware state/flags (e.g., `STC`, `HLT`).
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⚡ Quick Recall

8086 Instructions → Data (MOV) → Arithmetic (ADD/INC) → Logical (AND/SHL) → Branch (JMP/JZ) → String (MOVSB) → Control (STC/HLT)
